# Textbook Decoding Workspace: Reading a Reaction Coordinate Diagram for Enzyme Catalysis

**Course:** Introduction to Chemical Thinking-II/Methods in Chemical Biology

**Name:** _____ **Date:** _____

## Instructions

1. Open *Lehninger Principles of Biochemistry* to **Chapter 6 (Enzymes)** and locate **Figure 6-3** (the reaction coordinate diagram comparing an uncatalyzed reaction to an enzyme-catalyzed reaction).
2. Follow the steps across the horizontal axis from left (reactants) to right (products).
3. Complete every cell in the matrix below. Do not use external websites—your textbook is your sole authority.

## Part 1: The Visual-to-Conceptual Decoding Matrix

| **Graph Feature** | **1. Visual Audit** *(What exactly do you see?)* | **2. Molecular Reality** *(What are the molecules doing?)* | **3. The Text Verdict** *(Direct textbook definitions/quotes)* |
|---|---|---|---|
| **The Left Baseline** (Labeled **S**) | Is the initial flat line for **S** vertically higher or lower than the final flat line for **P** on the right?<br>☐ Higher<br>☐ Lower | What exact structural condition are the substrate molecules in right now before any kinetic collisions occur? | Find the bolded text surrounding Figure 6-3. Write down Lehninger's exact definition of the **Ground State**: |
| **The High Peak** (Labeled $\Delta G^{\ddagger}_{\text{uncat}}$) | Trace the dark, uncatalyzed line to its peak. Note the vertical arrow. Where does this arrow begin and end? | If a substrate molecule does not have an enzyme helper, what must happen to its chemical bonds to force it to this peak? | Quote the textbook sentence defining the **Transition State (‡)**. Why is this *not* a stable chemical species? |
| **The Dual Peaks & Valley** (Labeled **ES** / **EP**) | Look at the catalyzed line. How many mini-peaks does it have? What features sit in the shallow valley between them?<br>Peaks: _____<br>Valley: _____ | Describe what happens physically when the substrate drops into the active site. What are the enzyme and substrate forming? | Find the paragraph discussing reaction intermediates. Based on the reading, what do the abbreviations stand for?<br>**ES** = _____<br>**EP** = _____ |
| **The Peak Height Difference** (Comparing $\Delta G^{\ddagger}$) | Compare the vertical height of the uncatalyzed peak to the highest catalyzed peak. Which curve demands a smaller energy "climb"? | Does an enzyme work by giving the substrate an extra "kick" of heat energy, or by altering the pathway? Explain. | Look up the definition of **Activation Energy ($\Delta G^{\ddagger}$)**. According to the text, how does a lower activation energy change the *rate* of the reaction? |
| **The Overall Drop** (Labeled $\Delta G^{\circ}$) | Measure the vertical gap between the **S** baseline and the **P** baseline. Does this net distance change depending on which path is taken?<br>☐ Yes ☐ No | If a catalyst cannot change where the starting point or ending point sits, what can it *never* change about a chemical reaction? | Find the term **Standard Free-Energy Change ($\Delta G^{\circ}$)**. Quote the sentence explaining if an enzyme can alter reaction equilibrium: |

## Part 2: Deconstructing Common Misconceptions

### 1. The "Valley" Misconception

A common mistake introductory students make is looking at the intermediate valley (**ES**) and thinking, *"Because this valley sits higher up on the graph than the starting line S, the ES complex must be highly unstable and ready to explode."*

✎ Looking closely at the geometry of the graph lines and reviewing the text, explain why this assumption is false.

✎ How does the energy of the **ES** complex compare directly to the transition states flanking it on either side? Why is it a localized rest stop?

### 2. The Catalyst Limitation

Imagine you add a massive concentration of an incredibly efficient, perfectly engineered enzyme to a beaker containing a reaction that is highly thermodynamically unfavorable (meaning $\Delta G^{\circ}$ is a massive positive number, and the product line **P** sits way higher than **S**).

✎ Will this enzyme force the reaction to synthesize an abundance of products?

✎ Cite specific wording from Lehninger's text on the limits of catalysis to support your answer.